Test Diverse Users

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Test Diverse Users before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Build the Recommender

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Permissions and Human Review

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompt Experiments

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Test Diverse Users trace and record the first step you can explain without help.

Use W2: Plan **coverage** for Evaluating varied cases. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Test Diverse Users explanation against the supplied facts before discussing it.

Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Evaluate independently

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Test Diverse Users target to a trace and a justified conclusion.

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Test Diverse Users case and show the deciding evidence.

A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

Compare and investigate.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A recommender serves 6 of 6 easy-book requests but 1 of 4 advanced-book requests. Report both results and identify the next test priority.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 34 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Test Diverse Users adds to Transparent Book Recommender and keep the evidence.

Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Read denominators

> Read denominators. Apply the supplied Test Diverse Users facts and keep your reasoning.

A report says 8 of 10 tests pass, including 0 of 2 invalid-**input** cases. What is the important weakness?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Plan coverage

> Plan coverage. Apply the supplied Test Diverse Users facts and keep your reasoning.

Name five distinct categories for the chooser evaluation.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair reporting

Repair reporting. Apply the supplied Test Diverse Users facts and keep your reasoning.

A learner drops two failed cases and reports 8/8. What should be restored?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Evaluate independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Evaluate independently. Apply the supplied Test Diverse Users facts and keep your reasoning.

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Protect final evidence

Protect final evidence. Apply the supplied Test Diverse Users facts and keep your reasoning.

You fix the tie bug using the final tie case. What should the next evaluation report say?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Test Diverse Users

> A fictional recommender is tested on distinct ordinary, equality, invalid-input and tie cases. Explain the labelled step before checking the changed condition.

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

AI ACADEMY / AI-2/34

Disjoint evaluation categories

	Full passes	Cases run
Ordinary	4	4
Equality	1	3
Invalid	2	3
Ties	1	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/34

Test Diverse Users

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Total count

Passes 4+1+2+1=8; cases 4+3+3+2=12.

2. Overall rate

8/12≈66.7% full passes.

3. Find weak slices

Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.

4. Prioritise repair

Inspect failed equality traces and any invalid-input safety consequence, not only the total.

CHANGE ONE THING

Someone reports only the ordinary slice.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can these records establish performance for different user groups?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

1. Prioritise repair

2. Total count

3. Find weak slices

4. Overall rate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Prioritise repair Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Someone reports only the ordinary slice. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The claim becomes 4/4 on ordinary cases, not 100% across all tests.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Eight nonordinary cases and their failures have been omitted.

Repair a missing building block

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages..

Return to this week and explain what you changed.

Review Permissions and Human Review (ai-2/31). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?.

Return to this week and explain what you changed.

Review Prompt Experiments (ai-2/28). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters..

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

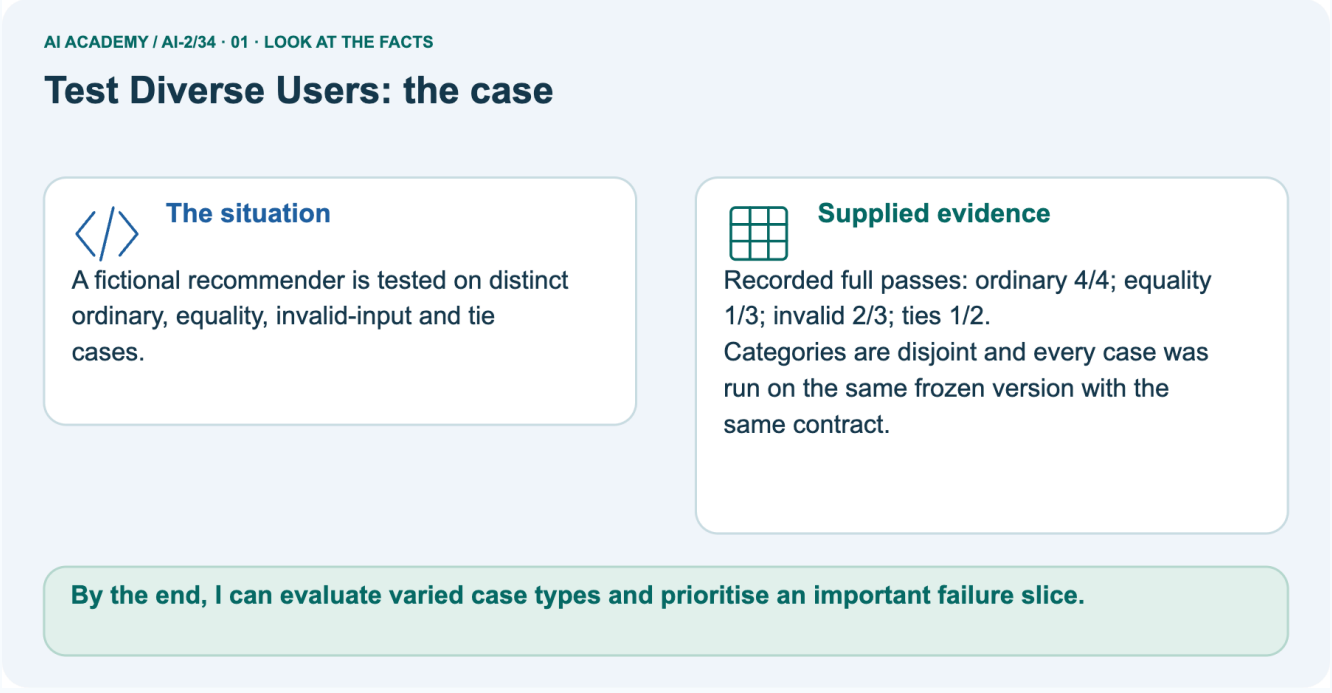
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Test Diverse Users picture from its labels, then record a relevant limit.

01. Test Diverse Users: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

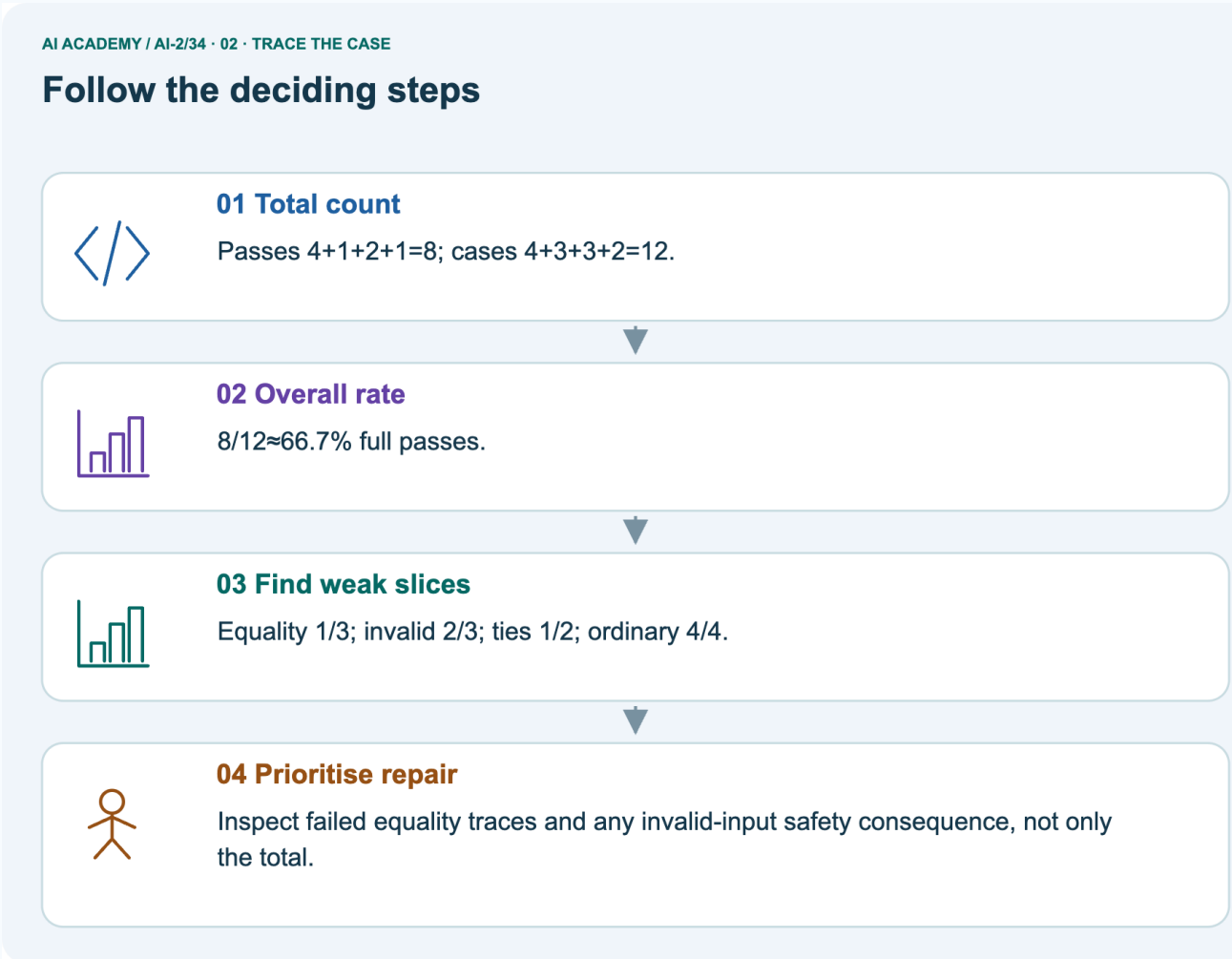
A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.
Supplied facts: Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



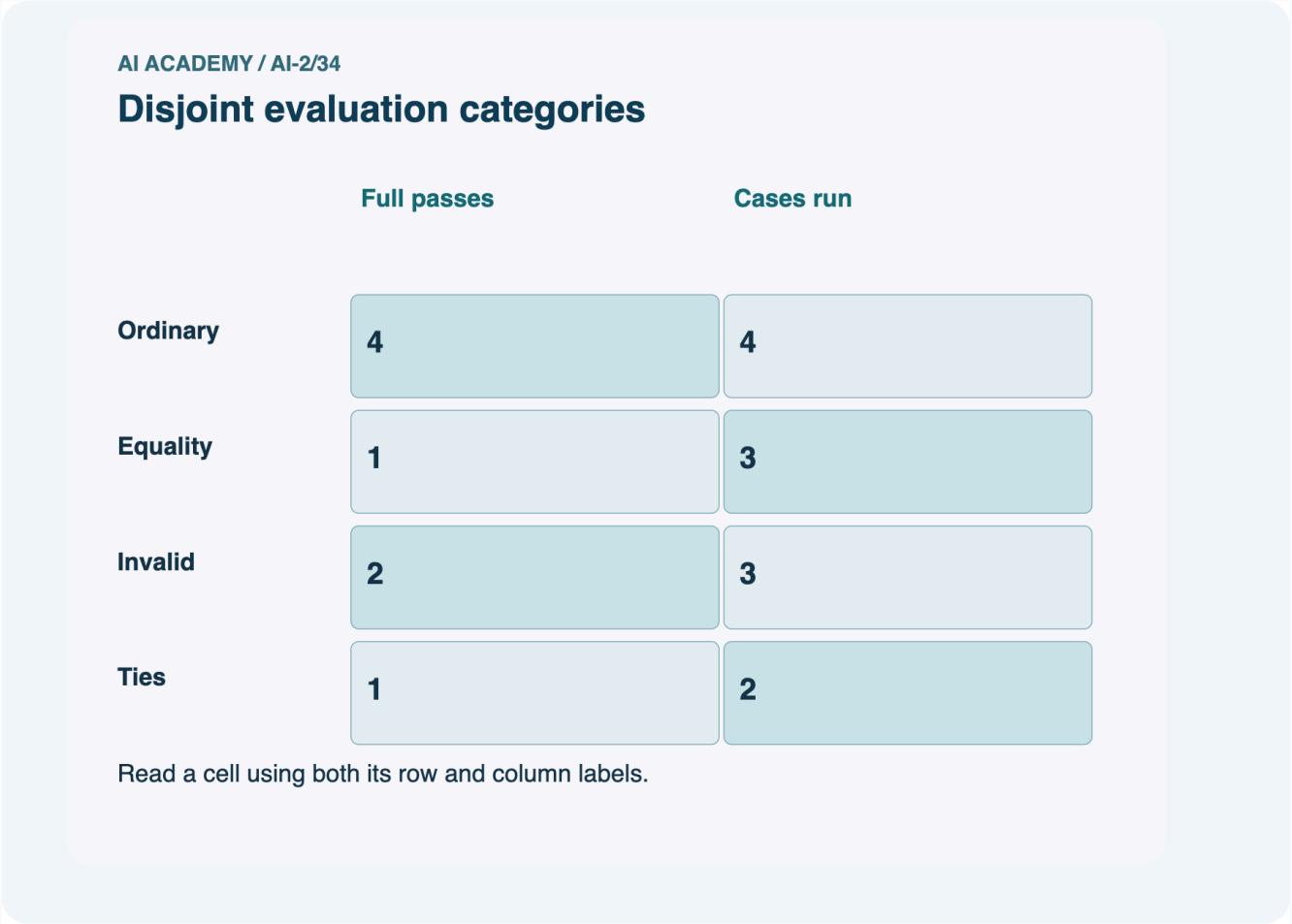
Follow the labelled picture with the matching explanation. Open the vector diagram

Total count: Passes 4+1+2+1=8; cases 4+3+3+2=12. → Overall rate: 8/12≈66.7% full passes. → Find weak slices: Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4. → Prioritise repair: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Disjoint evaluation categories



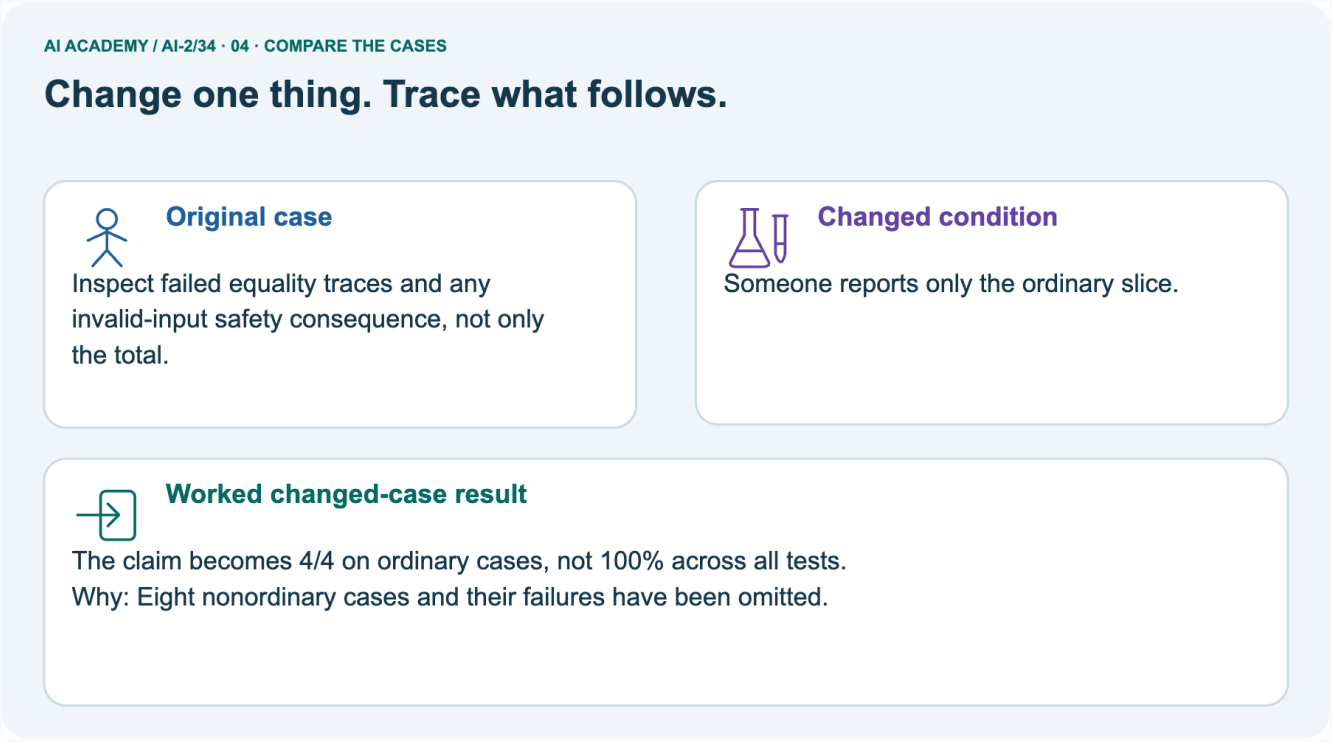
Follow the labelled picture with the matching explanation. Open the vector diagram

Disjoint evaluation categories. Ordinary: Full passes = 4, Cases run = 4. Equality: Full passes = 1, Cases run = 3. Invalid: Full passes = 2, Cases run = 3. Ties: Full passes = 1, Cases run = 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

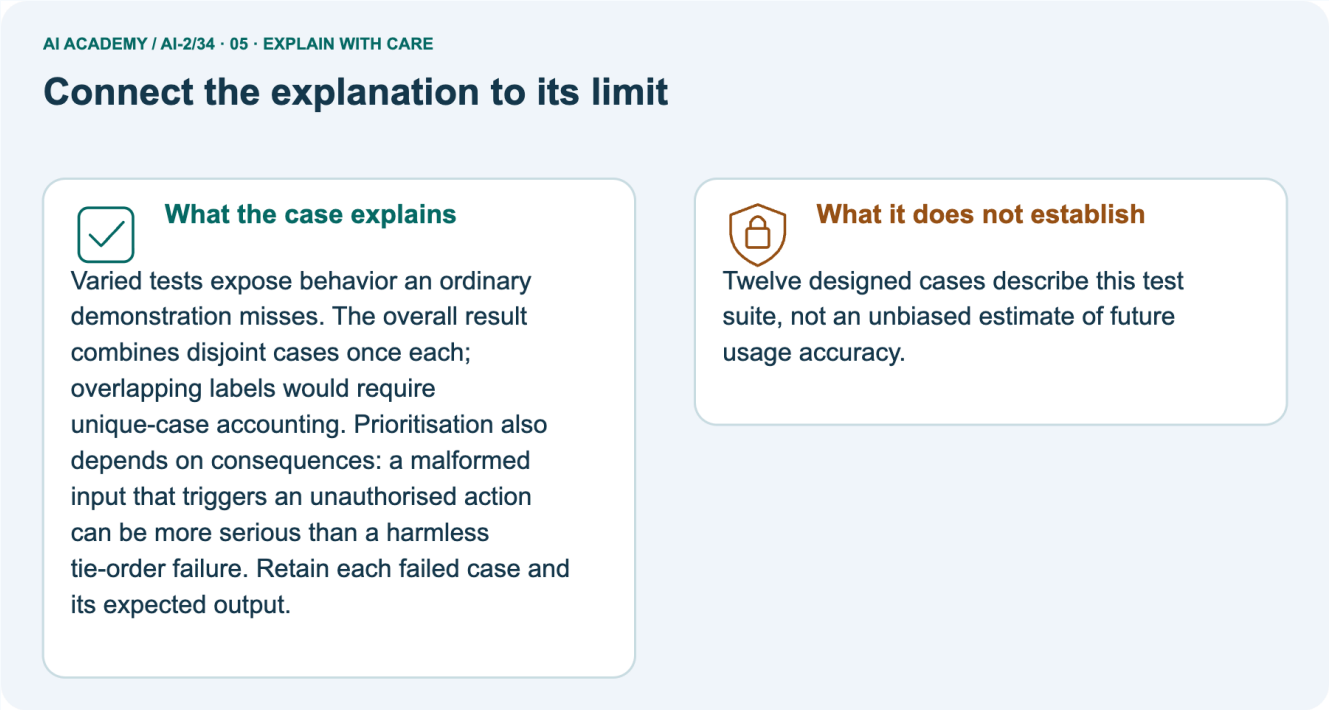
Original case: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total. Changed condition: Someone reports only the ordinary slice.

Expected result: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Reason: Eight nonordinary cases and their failures have been omitted.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

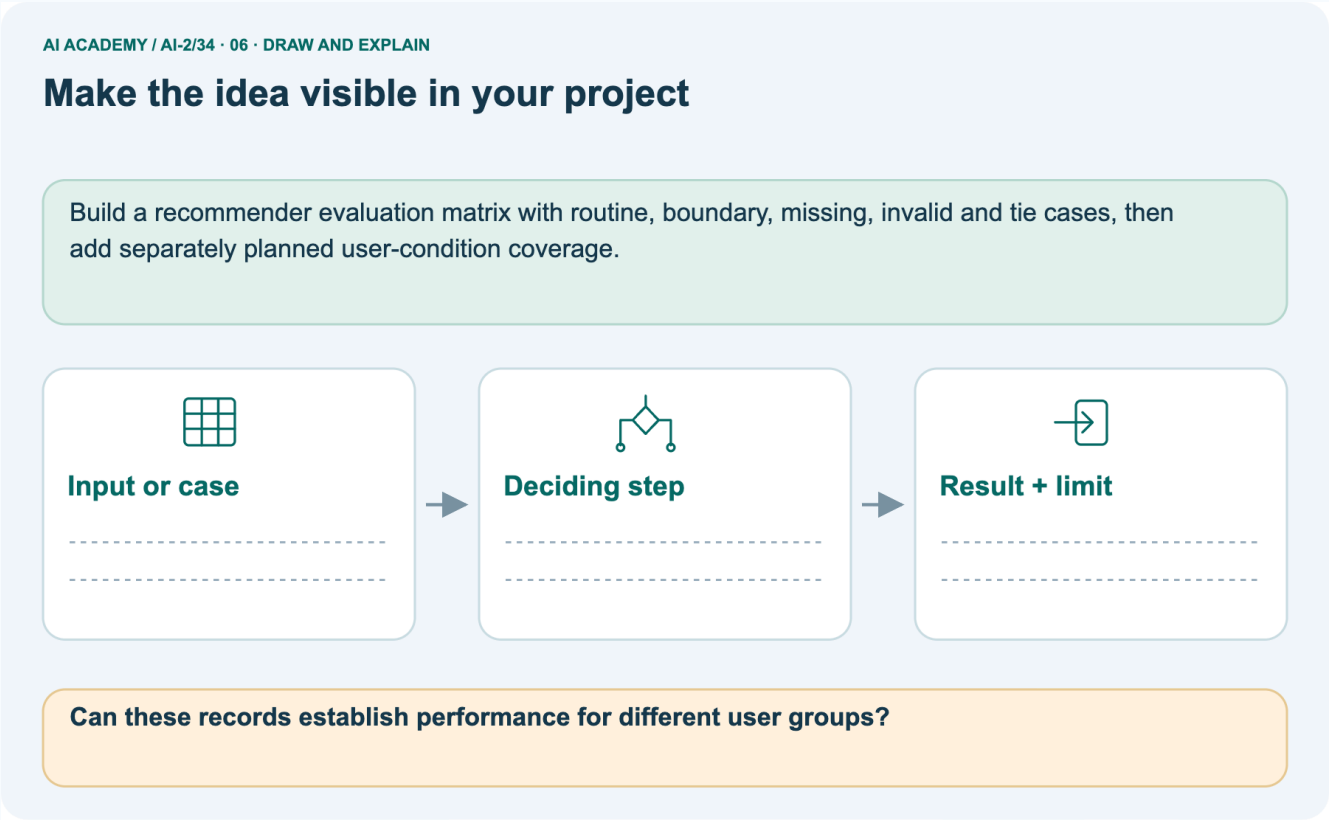
Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Limit: Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**. Draw your **input**, deciding step, result and limit.

Supported question: Can these records establish performance for different user groups?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.